REPORT ON Cross-dataset comparison Align APMS (Week 1) and CSEW (Week 2) on shared dimensions.

WEEK 2 ----- ANA 3

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1. Objective:

The main objective of this analysis was to perform a cross-dataset comparison between the APMS 2014 dataset and the CSEW dataset. The analysis focused on identifying similarities, differences, agreements, and conflicts between both datasets using shared mental health and social impact dimensions.

The comparison was based on prevalence indicators, demographic analysis, and visualization techniques. The purpose of this analysis was to better understand mental health treatment prevalence and domestic abuse related social impact trends within the population.

2. Datasets Used:

Two datasets were used in this analysis.

The first dataset was the APMS 2014 (Adult Psychiatric Morbidity Survey) dataset. From this dataset, Table 3.16 was used, which contained information related to treatment for mental or emotional problems by age and sex. The dataset included prevalence percentages for men, women, and all adults across different age groups.

	Age_Group	Men_Treatment	Women_Treatment
0	16-24	3.2	9.0
1	25-34	8.2	15.1
2	35-44	10.8	16.8
3	45-54	11.8	19.8
4	55-64	13.7	21.4
5	65-74	9.4	16.5
6	75+	9.6	15.7

The second dataset was the CSEW dataset (Crime Survey for England and Wales). From this dataset, Table S43c was used, which contained information related to domestic abuse prevalence trends. The dataset included indicators such as domestic abuse, partner abuse, family abuse, threats, and force-related prevalence percentages across multiple years.

	Indicator	2019-2020	2021-2022	2022-2023	2023-2024
0	Any domestic abuse (partner or family non-phys...	6.063285	5.727713	5.137845	5.351390
1	Any partner abuse (non-physical abuse, threats...	4.477245	3.923697	3.438816	3.553626
2	Any family abuse (non-physical abuse, threats,...	2.136765	2.436163	2.149497	2.122693
3	Partner abuse - non-sexual [note 14]	3.869533	3.223930	2.934945	3.115319
4	Non-physical abuse (emotional, financial) [not...	2.939518	2.548648	2.227238	2.581725
5	Threats or force [note 14]	2.009325	1.493146	1.464506	1.136618
6	Threats [note 14]	1.416103	1.213738	1.050243	0.753369
7	Force [note 14]	1.257405	0.686279	0.771066	0.681944
8	Family abuse - non-sexual [note 14]	1.849323	2.165337	1.899668	1.878842
9	Non-physical abuse (emotional, financial) [not...	1.246234	1.811761	1.401027	1.257299
10	Threats or force [note 14]	0.961539	1.099120	0.864622	0.907850
11	Threats [note 14]	0.749210	0.969015	0.726779	0.697595

3. Methodology:

The analysis was performed using Python libraries such as Pandas, NumPy, and Matplotlib within Google Colab.

Initially, the cleaned CSEW dataset was imported into the notebook for analysis. The APMS dataset values from Table 3.16 were manually extracted and organized for comparison purposes.

Several visualization techniques were used during the analysis process. These included bar charts and trend comparison graphs. Cross-dataset comparison was performed using shared dimensions such as prevalence percentage, gender analysis, trend analysis, and mental health related impact indicators.

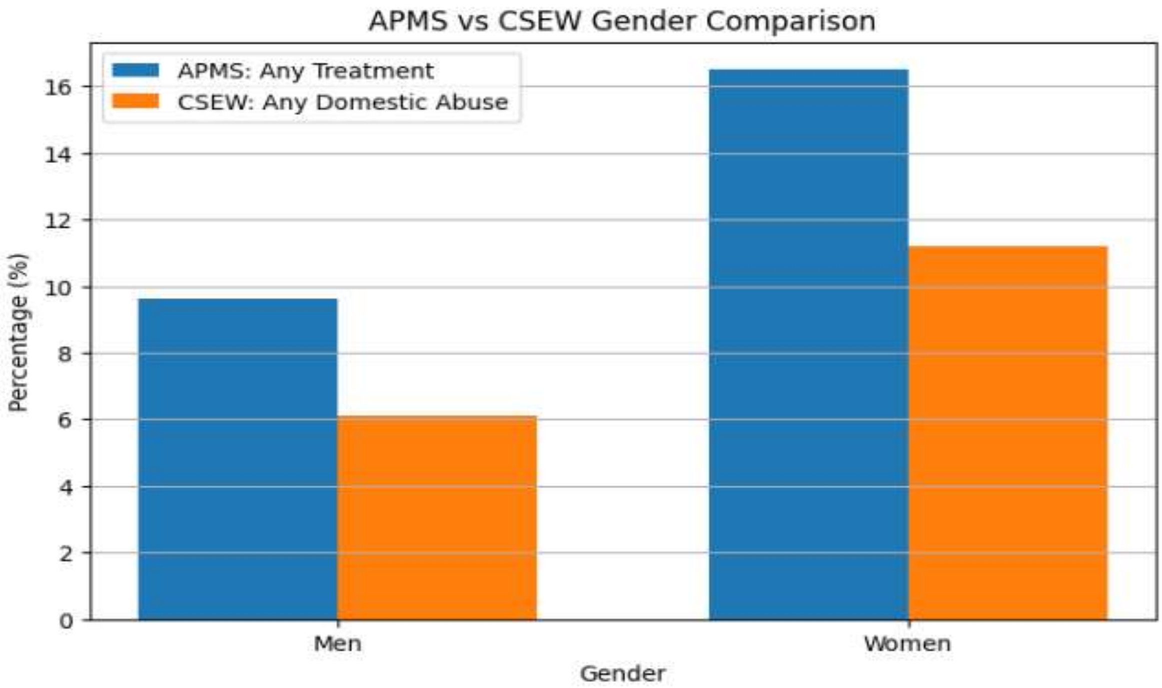
A comparison table was also created to summarize the similarities and differences between both datasets.

4. APMS vs CSEW Gender Comparison:

This graph compared APMS mental health treatment prevalence with CSEW domestic abuse prevalence across men and women.

The analysis showed that women had higher prevalence percentages compared to men in both datasets. In the APMS dataset, women showed higher treatment prevalence percentages than men. Similarly, the CSEW dataset also reflected higher prevalence percentages for women in abuse-related indicators.

This comparison highlighted the significant impact of mental health and social wellbeing issues on women within the population.

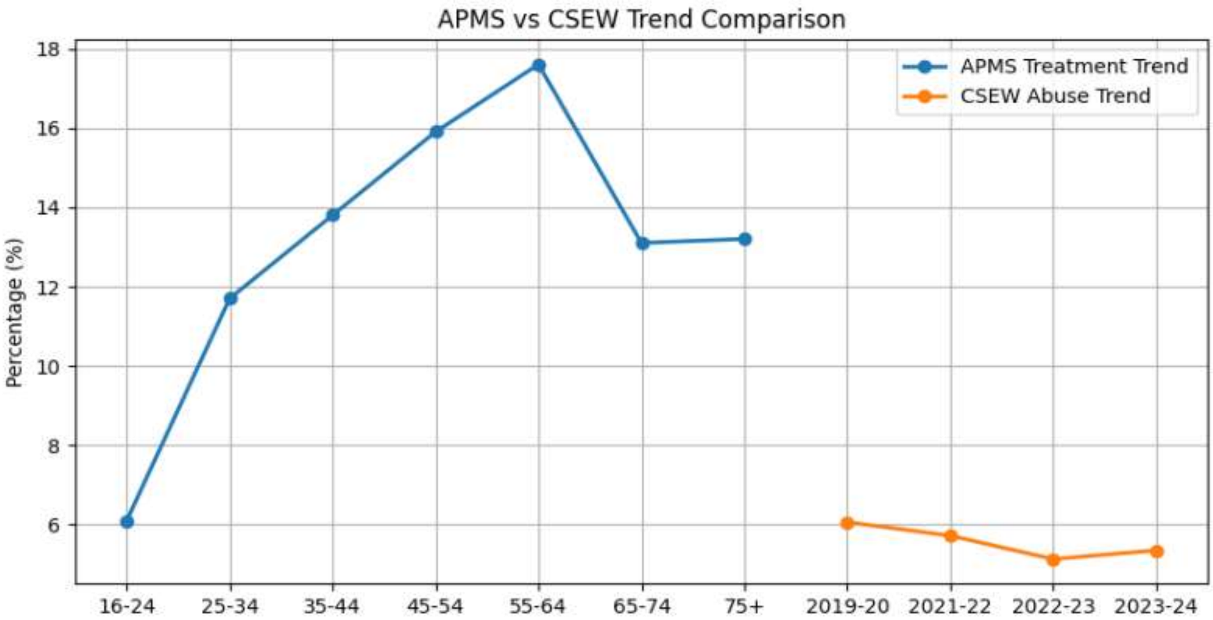


5. APMS vs CSEW Trend Comparison:

This graph compared APMS treatment prevalence trends with CSEW abuse prevalence trends.

The APMS dataset represented age-based treatment prevalence trends, while the CSEW dataset represented year-wise abuse prevalence trends. Although both datasets focused on different indicators, the comparison helped in understanding the variation of mental health and social impact related prevalence patterns.

The graph also demonstrated how visualization techniques can be used for identifying demographic and trend-based patterns across different datasets.



6. Cross-Dataset Comparison Table:

The comparison table summarized important similarities and differences between the APMS and CSEW datasets.

The APMS dataset mainly focused on mental health treatment prevalence and psychological wellbeing, whereas the CSEW dataset focused on domestic abuse prevalence and social impact indicators.

The comparison also showed differences in trend analysis techniques, where APMS used age-based prevalence analysis and CSEW used year-wise trend analysis.

	Dimension	APMS	CSEW
0	Dataset Focus	Mental health treatment	Domestic abuse prevalence
1	Indicator Used	Any treatment prevalence	Abuse prevalence trend
2	Trend Type	Age-based trend	Year-wise trend
3	Gender Analysis	Men vs Women	Men vs Women
4	Mental Health Impact	Psychological wellbeing	Social impact

7. Agreements:

Both the APMS 2014 dataset and the CSEW dataset focused on issues related to mental health, emotional wellbeing, and social impact within the population. Although the datasets were created for different purposes, both highlighted important public health and social concerns.

Both datasets used prevalence-based statistical analysis to measure the percentage of people affected by specific problems. APMS measured treatment prevalence for mental or emotional problems, whereas CSEW measured prevalence of domestic abuse and related social issues.

A major similarity between both datasets was the inclusion of demographic analysis. Both analyses considered gender-based comparison and showed that women were more affected compared to men. In the APMS dataset, women had higher treatment prevalence percentages, while in the CSEW dataset women showed higher abuse-related prevalence percentages.

Both datasets used trend analysis techniques to identify important patterns. APMS represented age-based treatment trends, whereas CSEW represented year-wise prevalence trends. These trends helped in understanding how mental health and social impact indicators varied across groups and time periods.

Visualization techniques such as bar charts and trend graphs were used in both analyses to improve understanding and interpretation of the data. These visualizations made demographic and prevalence comparisons more clear and effective.

Both datasets also highlighted the importance of mental health awareness, public wellbeing, and the need for improved support systems for affected individuals.

8. Conflicts:

Despite several similarities, both datasets differed significantly in terms of focus, methodology, and analysis objectives.
The APMS dataset primarily focused on mental health treatment prevalence and emotional wellbeing. It analysed how many people received treatment or counselling for mental or emotional problems across different age groups and genders.

On the other hand, the CSEW dataset mainly focused on domestic abuse prevalence, social safety issues, harassment, threats, and abuse-related indicators. Its objective was more related to crime, victimization, and social impact analysis.
The APMS dataset used age-based demographic analysis, while the CSEW dataset mainly used year-wise prevalence trend analysis. Therefore, the structure and interpretation of trends differed between both datasets.
Another important difference was the type of indicators used. APMS included treatment-related indicators such as medication, counselling, and emotional support, whereas CSEW included abuse-related indicators such as partner abuse, threats, force, and harassment prevalence.

The methodologies used for data collection and survey design were also different in both datasets. As a result, direct statistical comparison between the two datasets had certain limitations.

Finally, the APMS dataset mainly emphasized healthcare access and psychological wellbeing, whereas the CSEW dataset emphasized social impact, abuse prevalence, and public safety concerns.

9. Conclusion:

The cross-dataset comparison demonstrated that both APMS and CSEW datasets provide valuable insight into mental health and social wellbeing trends within the population.

The analysis showed that women had higher prevalence percentages compared to men in both datasets. This indicated the importance of mental health awareness, treatment accessibility, and social support systems.

The comparison also highlighted important similarities and differences in treatment prevalence, abuse prevalence, demographic analysis, and trend analysis patterns between the two datasets.

Overall, the analysis successfully identified agreements and conflicts between both datasets and demonstrated the usefulness of cross-dataset comparison techniques in data analytics and mental health related research.